GRAPHS

I've listed the broad topics and for every topic I've given the reading material, code and the questions. Do follow it in that order especially if you weren't able to attend the sessions. Also the questions taken up in the sessions were a bit on the difficult side so it's fine if you do those after the practice questions. Hope it helps!

PRE-REQS

- C/C++ Program for Graph and its representations
- Breadth First Search or BFS for a Graph
- Depth First Search or DFS for a Graph

TOPICS

1. BFS/DFS

CODE

- Traversals

PRACTICE QUESTIONS

- 841. Keys and Rooms
- 547. Number of Provinces
- 1020. Number of Enclaves
- Strengthen It
- Flood Fill
- Leetcode : Time Needed to Inform All Employees
- 542. 01 Matrix
- Shortest Path in Binary Matrix
- Leetcode : Shortest Path with Alternating Colors
- Word Ladder II
- 1162. As Far from Land as Possible
- 1293. Shortest Path in a Grid with Obstacles Elimination
  (Google Interview Question)

DONE IN SESSION

- 1306. Jump Game III
- Number of Islands
- Max Area of Island

- 1254. Number of Closed Islands
- 994. Rotting Oranges
- 130 Surrounded Regions
- Word Ladder

2. CYCLE DETECTION

SUPPLEMENTARY MATERIAL

- C/C++ Program for Detect cycle in an undirected graph
- Detect Cycle in a directed graph using colors
- Detect Cycle in a Directed Graph using BFS

CODE

- Cycle detection (undirected and directed)

PRACTICE QUESTIONS

- 802. Find Eventual Safe States

DONE IN SESSION

- 207. Course Schedule
- 1559. Detect Cycles in 2D Grid

3. TOPOLOGICAL SORT

SUPPLEMENTARY MATERIAL

- C/C++ Program for Topological Sorting
- Kahn's algorithm for Topological Sorting
- Detect Cycle in a Directed Graph using BFS
- All Topological Sorts of a Directed Acyclic Graph

CODE

- Topo sort - Using DFS and BFS (Kahn's algo)

PRACTICE QUESTIONS

- LeetCode 210. Course Schedule II
- 1462. Course Schedule IV
- Alien Dictionary | Practice | GeeksforGeeks

DONE IN SESSION

- Problem - 1388D
  (Solution - https://p.ip.fi/_GSk)

4. BIPARTITE GRAPH

SUPPLEMENTARY MATERIAL

- C/C++ Program for Check whether a given graph is Bipartite or not

PRACTICE QUESTIONS

- Leetcode : Possible Bipartition
- Is Graph Bipartite?

DONE IN SESSION

- Codeforces
  (Solution and code to check bipartite - https://p.ip.fi/kXj0)

5. DIJKSTRA'S ALGORITHM

SUPPLEMENTARY MATERIAL

- Dijkstra's Shortest Path Algorithm using priority_queue of STL

PRACTICE QUESTIONS

- 1514. Path with Maximum Probability
- Leetcode 743 Network Delay Time
- Shortest in shortest

DONE IN SESSION

- 1631. Path With Minimum Effort
- Problem - 938D (Codenation and Media.net question)
  (Solution - https://p.ip.fi/KXBr)

6. MINIMUM SPANNING TREE

SUPPLEMENTARY MATERIAL

- Minimum Spanning Tree Tutorials & Notes | Algorithms

CODE

- Prim's Algorithm
- Kruskal + DSU

You can practice coding up the 2 algos and run the code in the IDE given at the end of the supplementary material.

7. DISJOINT SET UNION

SUPPLEMENTARY MATERIAL

- Disjoint Set Union - Competitive Programming Algorithms
  (Naive Implementation, Path compression, Union by size and rank, Time Complexity)

PRACTICE QUESTIONS

- Redundant Connection
- Redundant Connection II
- [947] Most Stones Removed with Same Row or Column
  (Google Interview question)
- 1319. Number of Operations to Make Network Connected
- 990. Satisfiability of Equality Equations
- 721. Accounts Merge

DONE IN SESSION

- HELP Maximum Subtree
  (Solution - https://p.ip.fi/1qfP)

8. FLOYD WARSHALL

SUPPLEMENTARY MATERIAL

- Floyd-Warshall Algorithm

PRACTICE QUESTIONS

- Find the City With the Smallest Number of Neighbors at a Threshold Distance

DONE IN SESSION

- Friends - Kick Start
  (Solution - https://p.ip.fi/FIcM)

9. RE-ROOTING DP

SUPPLEMENTARY MATERIAL

- Tree Rerooting Technique in C++

PRACTICE QUESTIONS

- Tree Distances I
- F

DONE IN SESSION

- 834. Sum of Distances in Tree
- Problem - 1324F
  (Solution - https://p.ip.fi/e4o4)

10. KOSARAJU'S ALGORITHM (STRONGLY CONNECTED COMPONENTS)

SUPPLEMENTARY MATERIAL

- Strongly Connected Components and Condensation graph - Competitive Programming Algorithms

DONE IN SESSION

- Problem - 427C
  (Solution and code for kosaraju - https://p.ip.fi/fBmv)

11. BELLMAN FORD (NEGATIVE CYCLE DETECTION AND SHORTEST PATH)

SUPPLEMENTARY MATERIAL

- Bellman–Ford Algorithm | DP-23